

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: ASAP++ | Context: Qatari and Arab High School Arabic Essay Writers (Grades 10–12)

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's category inventory consists of three essay types — argumentative/persuasive, source-dependent, and narrative/descriptive [\[Q12, Q13, Q14, Q15\]](#) — drawn from a US middle-school context. The deployment requires coverage of literary analysis, religious-text commentary, and cultural commentary genres that are standard in Arab high school curricula but entirely absent from the benchmark. The Qatar MOEHE curriculum confirms Arabic Literature as a distinct subject and Islamic Education as compulsory [\[WEB-11, WEB-5\]](#), yet no AES benchmark covers these genres in Arabic. The expert elicitation flagged IO as HIGH priority precisely because the benchmark's taxonomy omits the test-case types the deployment requires. Even within the persuasive genre that nominally overlaps, scoring logic is type-specific and embedded in English pedagogical norms [\[Q43, Q44, Q45\]](#).

ID	Evidence
Q12	"There are 3 types of essays in the dataset." (p.2)
Q13	"Argumentative / Persuasive essays - These are essays where the prompt is one in which the writer has to convince the reader about their stance for or against a topic (for example, free speech in public colleges)." (p.2)
Q14	"Source-dependent responses - These essays are responses to a source text, where the writer responds to a question about the text (for instance, describing the writer's opinion about an incident that happened to him in the text)." (p.2)
Q15	"Narrative / Descriptive essays - These are essays where the prompt requires us to describe / narrate a story." (p.2)
Q43	"For source-dependent essays, we found out that the most important feature for content was length, while for argumentative / persuasive essays, it was coherence and cohesion features, followed by length." (p.4)
Q44	"For source-dependent essays, the coherence and cohesion feature set is the most important feature set for each of the other 3 attributes." (p.4)
Q45	"For persuasive / argumentative essays, coherence and cohesion features are the most important features for 4 of the 5 attributes." (p.4)
WEB-5	Islamic Education is compulsory in Qatar national curriculum schools for Qatari pupils and Arab passport holders (https://www.qisweb.qis.org/arabic-department/)
WEB-11	IMPACT-se review of Qatar 2025–2026 curriculum confirms Arabic Literature and politically/religiously contextualized content (https://www.impact-se.org/wp-content/uploads/Qatar-Report-2026-final.pdf)

Strengths

- The conceptual move toward attribute-level (multi-trait) scoring across multiple essay types [\[Q4, Q18\]](#) is methodologically analogous to what an Arabic formative feedback system would target, as later Arabic AES work like LAILA's seven-trait rubric demonstrates [\[WEB-12\]](#).

ID	Evidence
Q4	"Our contribution is the scoring of individual attributes of the essays, like content, organization, style, etc. in the ASAP dataset for the remainder of the essays." (p.1)
Q18	"Based on the types of essays, there are 2 sets of attributes." (p.2)
WEB-12	LAILA used annotators trained on Arab high school writing — example of appropriate annotator profile for this deployment (https://arxiv.org/abs/2512.24235)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: The deployment requires at least five genres: persuasive/argumentative (ةيعانقإ ةلاقم), explanatory/expository, literary analysis (ةلأق ليلحت ةلاقم), religious-text commentary (ريسفت / ريدي صن لمأت), and cultural commentary (ةلاقم ةلفاقت / ةيعامتج), grounded in Qatar MOEHE Arabic Language and Arabic Literature curricula [WEB-10, WEB-11, WEB-5].

IO-2: Yes — literary analysis, religious-text commentary, and cultural commentary are all absent from the benchmark, which covers only argumentative/persuasive, source-dependent, and narrative/descriptive types [Q12, Q13, Q14, Q15].

IO-3: The narrative/descriptive category and source-dependent responses keyed to specific US source texts [Q14, Q15] introduce categories not aligned with formal Arab high school assessment, where literary analysis and religious-text engagement are more central [WEB-11].

IO-4: Category gaps — absence of literary analysis, religious commentary, and cultural commentary — represent severe construct underrepresentation. The benchmark also embeds genre-specific scoring rules (length-weighted for source-dependent; coherence/cohesion-weighted for argumentative) [Q43, Q44, Q45] that would not transfer to Arabic rhetorical norms even within the nominally overlapping argumentative genre.

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WEB-5	Islamic Education is compulsory in Qatar national curriculum schools for Qatari pupils and Arab passport holders (https://www.qisweb.qis.org/arabic-department/)
WEB-10	Qatar MOE Arabic Language curriculum framework anchors Grades 1–12 writing standards (http://www.sec.gov.qa/En/Education/Pages/GradeAndSubject.aspx)
WEB-11	IMPACT-se review of Qatar 2025–2026 curriculum confirms Arabic Literature and politically/religiously contextualized content (https://www.impact-se.org/wp-content/uploads/Qatar-Report-2026-final.pdf)
WEB-12	LAILA used annotators trained on Arab high school writing — example of appropriate annotator profile for this deployment (https://arxiv.org/abs/2512.24235)

Information Gaps

- Whether Quranic/religious-text commentary essays are graded within the Arabic Language curriculum versus only Islamic Education is not resolvable from open sources.

- Cross-country curricular variation among Arab nations for genre coverage and weighting is undocumented in open sources.

Requires Expert Verification

- Genre inventory and weighting in Qatar MOEHE Grades 10–12 Arabic writing assessment
- Whether literary analysis essays follow distinct rubric structures across Arab national curricula

Remediation

Gap 1: Benchmark covers only argumentative/persuasive, source-dependent, and narrative/descriptive genres; deployment requires literary analysis, religious-text commentary, and cultural commentary.

Recommendation: Construct a genre inventory aligned with Qatar MOEHE Arabic Language and Arabic Literature curricula [WEB-10, WEB-11], using LAILA's prompt set [WEB-12] as a partial proxy and commissioning new Qatari teacher-authored prompts for the three missing genres.

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